

CampusConnect

Comprehensive Event Management System - Interview Preparation Guide

1. Project Vision & Purpose

CampusConnect is an enterprise-grade web application designed to digitize and streamline the lifecycle of campus events. It bridges the gap between **Club Administrators**, **Super Admins**, and **Students**.

Key Achievement: Successfully automated registration for over 1,000 students while providing real-time data sync to Excel for offline event management.

2. Technology Stack Architecture

Layer	Technology	Why this choice?
Frontend	React 19 + Vite	Modern UI, extremely fast bundling, and superior state management for complex forms.
Backend	Node.js + Express	Scalable, event-driven architecture that handles concurrent registrations efficiently.
Database	MySQL	Relational integrity ensures that student data, club roles, and event schedules are perfectly linked.
Security	JWT + Bcrypt	Stateless authentication for security and hashed passwords for data privacy.
Services	Multer + Nodemailer	Handles image uploads and automated entry-pass dispatch via email.

3. Advanced System Design Patterns

During the interview, mention these to demonstrate senior-level thinking:

Singleton Pattern

Implemented in `db.js` for the MySQL connection pool and `api.js` for the Axios instance. This ensures global consistency and prevents resource exhaustion.

Idempotency

The registration logic checks for existing records before processing. This prevents "Double-Registration" bugs if a user submits a form multiple times due to slow internet.

DAO Pattern

Our models (like `registrationModel.js`) use static methods to encapsulate all database logic, keeping the controllers clean and focused only on business logic.

Factory Pattern (Proposed)

"In the future, I plan to implement a Notification Factory to handle different types of alerts (Email, SMS, Push) dynamically based on event priority."

4. Development Phases

1. **Requirement Engineering:** Mapping user roles (Admin vs Student) and defining core features.
2. **Database Schema Design:** Normalizing tables to ensure data consistency across events and users.
3. **RESTful API Development:** Building 20+ secure endpoints with proper HTTP status codes.
4. **Frontend Integration:** Building a responsive dashboard with Tailwind CSS and Framer Motion.
5. **Business Intelligence:** Developing the Excel export engine for administrative reporting.
6. **Deployment & UAT:** Simulated testing with bulk data to verify system stability.

5. Interview Cheat Sheet

"What was the most challenging part?"

Answer: "Synchronizing real-time database registrations with a downloadable Excel file. I solved this by implementing a background sync task that ensures the file is updated every time a new student registers without blocking the main thread."

"How did you handle security?"

Answer: "I implemented a layered security approach: Bcrypt for password hashing, JWT for session security, and custom Express middleware to enforce Role-Based Access Control (RBAC)."

Prepared for Interview Success • CampusConnect Project Documentation

© 2026 CampusConnect Terminal